Probabilistic Dataset Merge

Scenario 2: Rounding Discrepancy

What happens when the target uses coarser precision than the supplemental dataset

What this scenario demonstrates. A common real-world problem: the same data appears at different levels of precision in different sources. Here the target dataset rounds population counts to the nearest 100, ages to the nearest year, and percentages to the nearest whole number. The supplemental dataset retains the original decimal precision. The two records describe the same real-world unit — but because they report values at different precisions, the system cannot find an exact match. The correct row is still selected as the best match, but the distance is non-zero and the signals reveal the imprecision introduced by rounding.

The Target Record

This is the record we are trying to match. The system will search the supplemental dataset for the row closest to these values.

Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units
2,500	40.0	100.0	600

Table 1: Target row values (raw, unstandardized).

Note: *The target values above have been rounded from their original precision. Original values: Dragon Sightings = 2,469; Avg. Wizard Age = 40.2; Pct. Leprechaun Cottages = 99.5; Goblin Family Units = 649. The supplemental dataset retains this original precision — see rank 1 in the candidate table.*

All Candidate Matches

The table below shows all 20 supplemental rows, sorted from closest to farthest. Distances are computed in standardized space (see Section 3). The highlighted row is the best match — the one the system selects.

Rank	Dragon Sightings	Avg. Wizard Age	Pct. Leprechaun Cottages	Goblin Family Units	Distance
1	2,469	40.2	99.5	649	0.2902
2	2,948	33.4	98.5	687	1.1684
3	2,524	30.0	99.5	557	1.2327
4	3,885	37.2	100.0	1,004	1.5321
5	3,849	42.2	99.5	1,043	1.6060
6	4,158	29.8	99.4	1,089	2.1987
7	4,555	26.5	99.4	930	2.3809
8	2,512	25.8	96.1	521	2.6202
9	5,094	27.3	100.0	1,041	2.6813
10	4,822	25.7	98.2	914	2.6886
11	4,891	34.4	99.6	1,339	2.7606
12	5,006	30.1	97.5	1,230	3.0420
13	5,296	25.4	99.3	1,128	3.0439
14	4,223	36.3	94.2	1,132	3.5420
15	4,701	58.7	100.0	1,519	3.7521
16	6,303	29.8	100.0	1,490	3.8598
17	6,516	23.3	99.3	1,385	4.1215
18	3,465	47.9	92.5	1,080	4.1998
19	6,668	32.0	98.5	1,641	4.3168
20	7,389	35.0	100.0	1,851	4.9514

Note: "Distance" is the standardized Euclidean distance between the target and each candidate row. A distance of 0.0000 means a perfect, identical match.

How Distance Is Calculated: A Worked Example

We use the **rank 2** candidate below to illustrate the calculation step by step. (The rank 1 match has distance 0 and is trivial; rank 2 shows the full arithmetic.)

Step 1 — Put every feature on the same scale (standardization).

Raw values are measured in different units (counts, years, percentages). Before computing distance we convert each to a *standardized score*: how many standard deviations above or below the combined average that value sits.

$$z = \frac{\text{raw value} - \text{combined average}}{\text{combined spread}}$$

Feature	Target (raw)	Candidate (raw)	Combined Avg.	Combined Spread
Dragon Sightings	2,500	2,948	4465.43	1419.86
Avg. Wizard Age	40.0	33.4	33.86	8.33
Pct. Leprechaun Cottages	100.0	98.5	98.62	1.97
Goblin Family Units	600	687	1087.14	356.70

Applying the formula above to each feature gives these standardized scores:

Feature	Target (std.)	Candidate (std.)
Dragon Sightings	-1.3842	-1.0687
Avg. Wizard Age	+0.7373	-0.0549
Pct. Leprechaun Cottages	+0.7003	-0.0604
Goblin Family Units	-1.3657	-1.1218

Step 2 — Compute the squared difference for each feature, then sum and take the square root.

$$d = \sqrt{\sum_i \left(z_{\text{target},i} - z_{\text{cand},i}\right)^2}$$

Feature	Target (std.)	Cand. (std.)	Squared difference
Dragon Sightings	-1.3842	-1.0687	0.0996
Avg. Wizard Age	+0.7373	-0.0549	0.6275
Pct. Leprechaun Cottages	+0.7003	-0.0604	0.5787
Goblin Family Units	-1.3657	-1.1218	0.0595
Sum			*1.3652*

$$d = \sqrt{1.3652} = \mathbf{1.1684}$$

Distance Distribution

The chart below shows the distance from the target to each of the 20 candidate rows, sorted closest to farthest. The **highlighted bar** is the selected match.

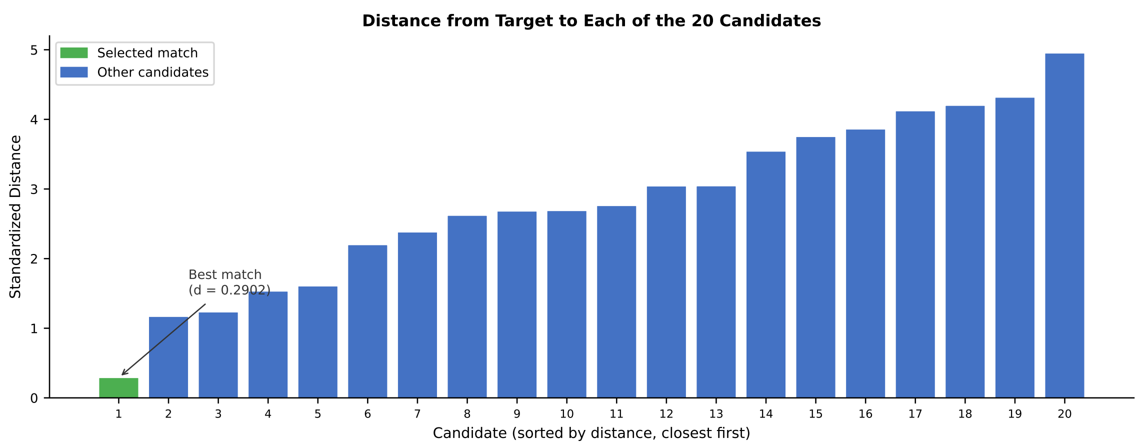

Figure 1: Distances from target to all 20 supplemental candidates, sorted ascending. The selected match is highlighted.

Signal Analysis

Each signal below is reported in the output file for every matched row. This section explains what each signal means and why it has the value it does for this particular match.

euc_distance `0.2902`

The best match distance is **0.2902** — non-zero despite the correct row existing in the supplemental dataset. The gap comes entirely from rounding: the target reports rounded values while the supplemental retains the original precision. In the exact match scenario this was 0.0000; here the same underlying record produces a small but real distance.

per_feature_contribution (best match)

Shows how much of the best-match distance each feature is responsible for. In a rounding scenario the contribution concentrates in whichever features were rounded most aggressively relative to their spread — here a large share can land on a single percentage column simply because rounding 99.5% to 100% is a big move in standardized units. **Why is a high share not flagged?** No flag rule fires on contribution alone — it is a diagnostic, not a verdict. Concentration only signals trouble when it is paired with a **large** absolute distance (see Scenario 3); here the total distance is tiny, so a concentrated share just means the other features agree almost perfectly.

Feature	Share of distance
Dragon Sightings	0.6%
Avg. Wizard Age	0.7%
Pct. Leprechaun Cottages	76.3%
Goblin Family Units	22.4%

nndr `0.2484`

NNDR $= d_1/d_2 = 0.2902 \div 1.1684 = \mathbf{0.2484}$. This is below the 0.80 threshold, so no ambiguity flag is raised — the system is still confident the best match is correct. However, note that the NNDR is no longer 0.0000 as in the exact match: rounding has brought the second-best candidate slightly closer relative to the best match.

near_miss_count `0`

0 supplemental row(s) fall within the near-miss threshold ($d_1/d_i \geq 0.80$). Because the best-match distance is small but non-zero, the ratio d_1/d_i is also non-zero for all other rows — the cascading check stops quickly and no near misses are counted.

mnn_confirmed `True`

Confirmed: True. Running the search in reverse — from the matched supplemental row back to the target — still returns this target as the nearest record. Even with rounded target values, the match is symmetric.

repeats `1`

No exact tie — a single row sits alone at the minimum distance (the count includes the chosen match itself, so 1 means a unique winner).

smd (per feature) `[0.0000, 0.0000, 0.0000, 0.0000]`

With only one target row, the SMD cannot be computed and is reported as 0 for all features. In a full run, systematic rounding of the target dataset would appear here as a consistent offset on the affected features.

flags `(none)`

Flags: (none). No flags raised. Despite the non-zero distance, the system is confident enough in the match that no warnings are triggered. The distance and NNDR are visible in the output for the researcher to inspect.

Data note: Values in this example are adapted from U.S. Census Bureau American Community Survey (ACS) public data. Column names have been replaced with illustrative labels for demonstration purposes. This document is intended for researcher education and internal system validation.